

pyDMS

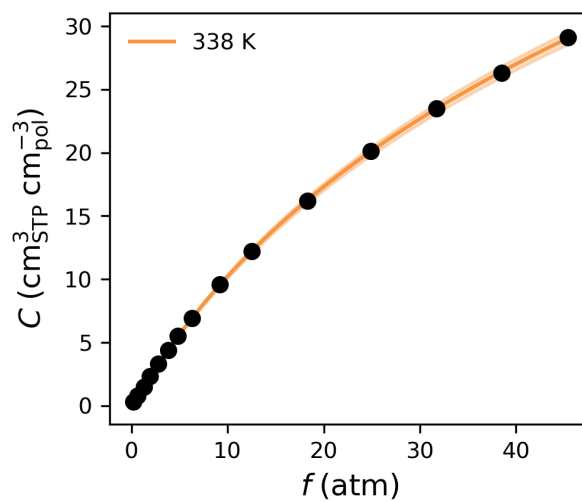
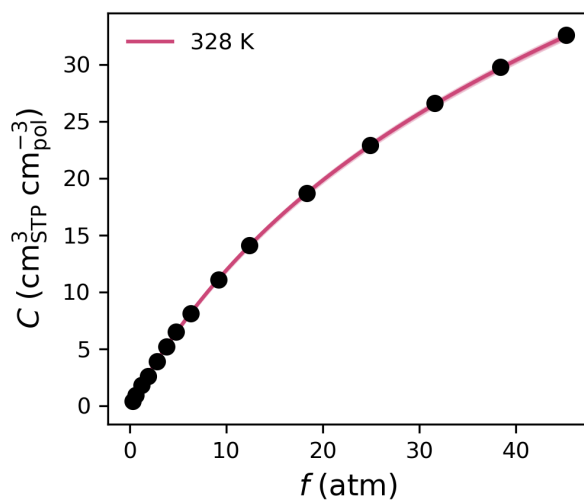
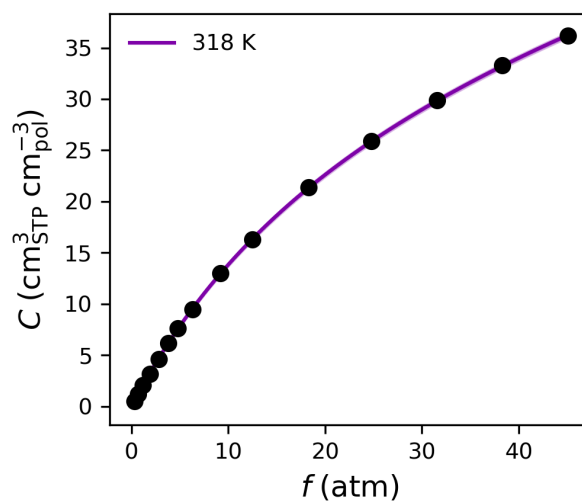
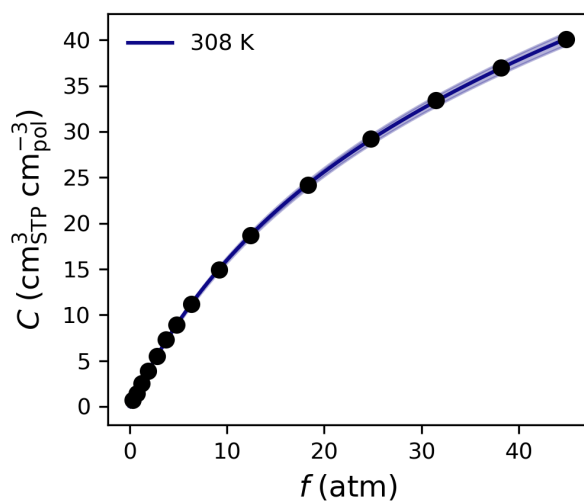
Reproducible Dual-Mode Sorption Parameters

Analysis performed at 2026-04-05 21:21:25 Eastern Daylight Time

DMS Parameters

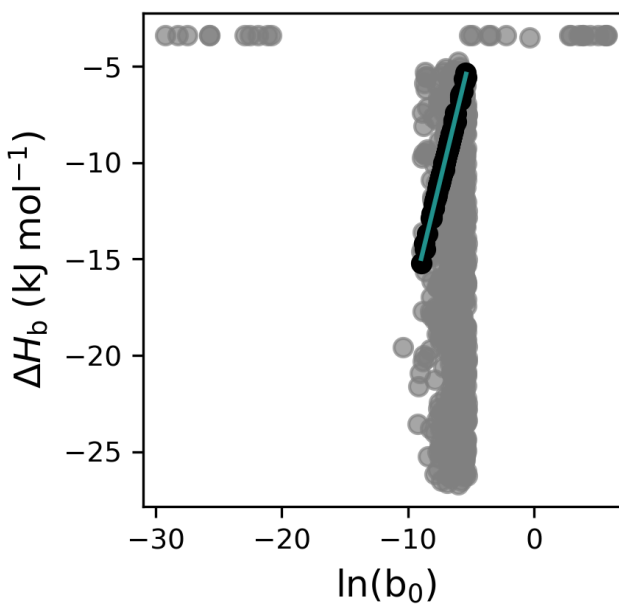
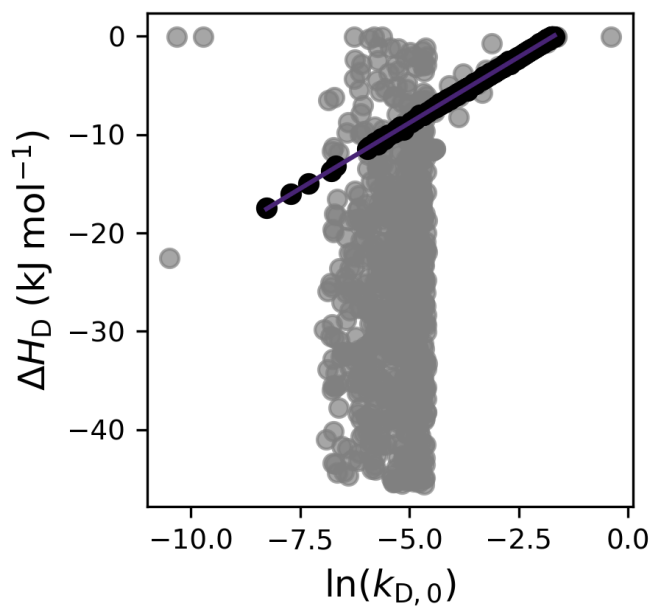
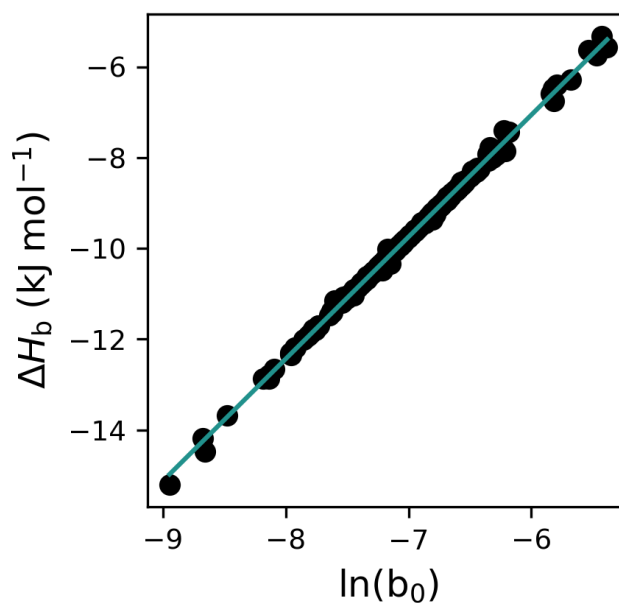
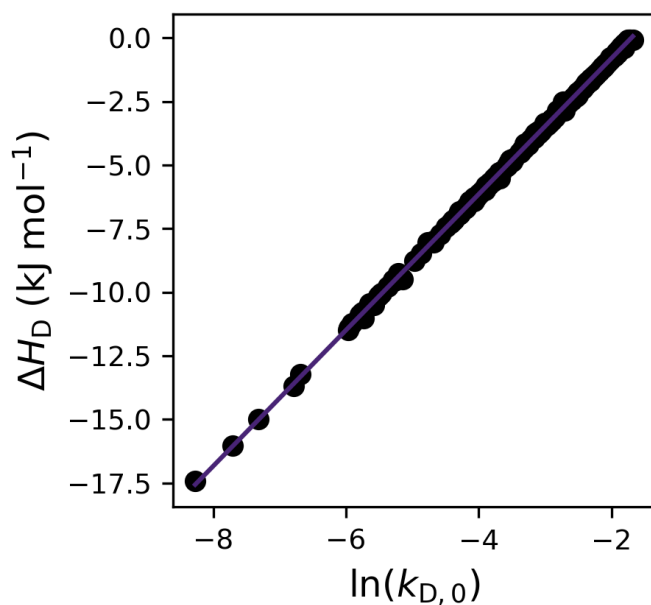
Temperature	Parameter	Result $\pm$ Std. Dev.
308 K	k_D	0.22369 ± 0.01836
	C'_H	45.60164 ± 1.61701
	b	0.04320 ± 0.00121
318 K	k_D	0.19202 ± 0.00431
	C'_H	44.09886 ± 0.49062
	b	0.03703 ± 0.00096
328 K	k_D	0.16637 ± 0.00606
	C'_H	42.25753 ± 0.53684
	b	0.03205 ± 0.00083
338 K	k_D	0.14538 ± 0.01301
	C'_H	40.12328 ± 1.43028
	b	0.02798 ± 0.00077

Sorption Isotherms

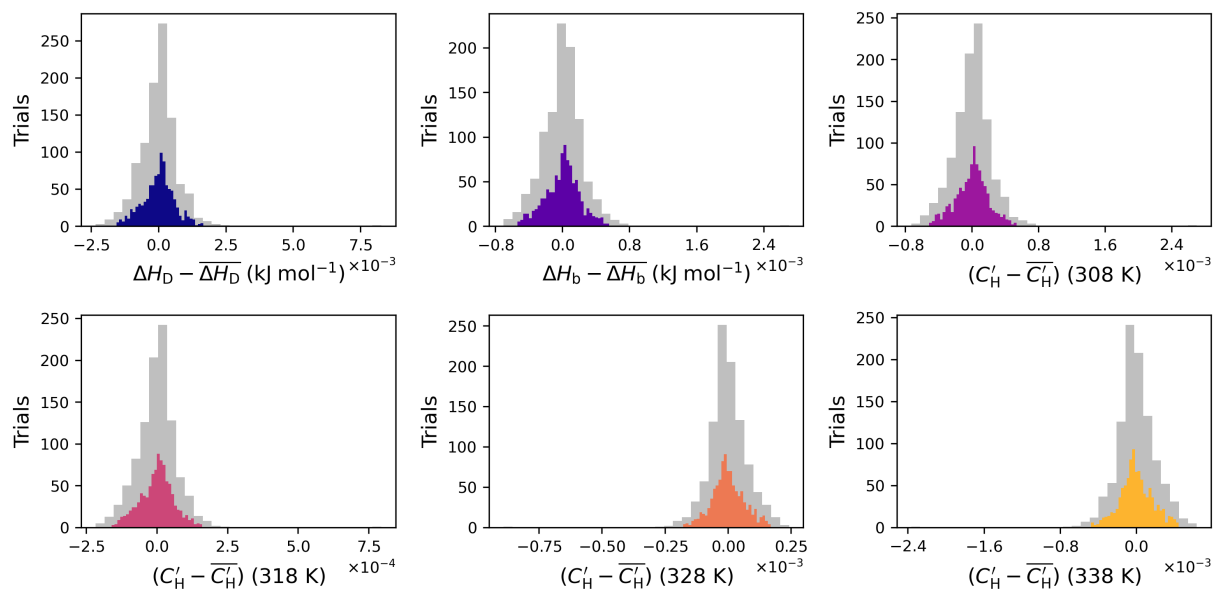


Linear Free Energy Relationships

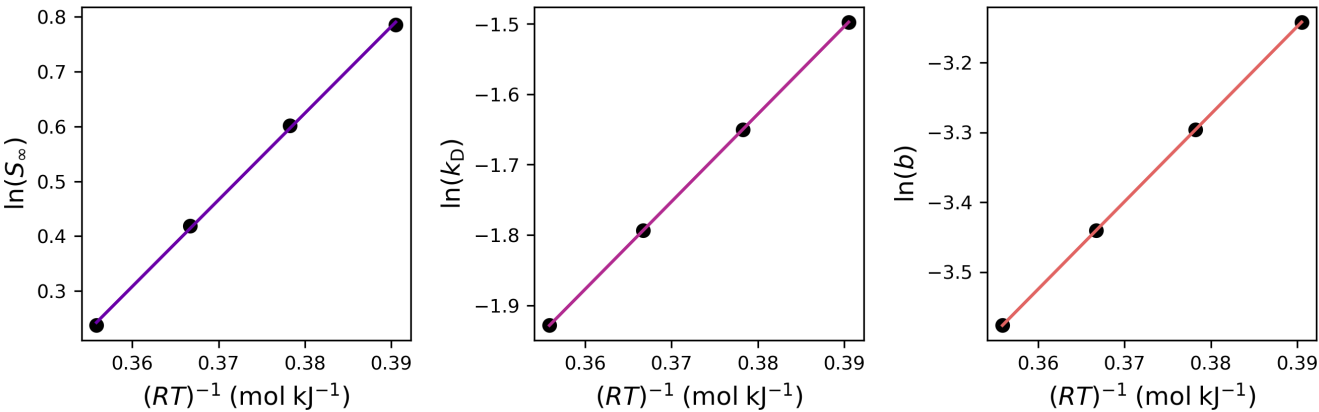
Heat of Henry Sorption (kJ/mol)	$\Delta H_D = 4.5558 + 2.6742 \ln(k_{D,0})$
Heat of Langmuir Sorption (kJ/mol)	$\Delta H_b = 9.0824 + 2.6897 \ln(b_0)$



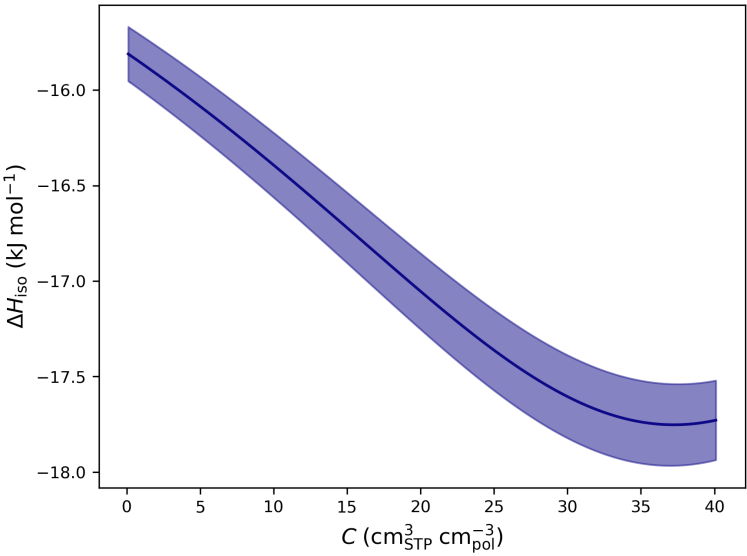
van't Hoff Relationships



Sorption Energetics



	Entropic Prefactor ($S_{i,0}$)	ΔH_i (kJ/mol)
i = Infinite Dilution	$4.60\text{e-}03\pm4.85\text{e-}04$	-15.8058 ± 0.2829
i = Henry	$1.74\text{e-}03 \pm 1.43\text{e-}16$	-12.4324 ± 4.9319
i = Langmuir	$3.23\text{e-}04 \pm 1.41\text{e-}17$	-12.5341 ± 0.6098



Analysis Settings

Parameter	Value
ΔH_D Initial Guess Range	[-45.60441789 -1.]
ΔH_b Initial Guess Range	[-26.7239349 -5.0912392]
$k_{D,0}$ Initial Guess Range	[0.001, 0.01]
b_0 Initial Guess Range	[0.0001, 0.005]
ΔH_D Solver Bounds	[-54.72530147 -0.0634579]
ΔH_b Solver Bounds	[-32.06872189 -3.39415947]
$k_{D,0}$ Solver Bounds	[0, None]
b_0 Solver Bounds	[0, None]
C'_H guess #1	[3.99077248 59.53602181]
C'_H guess #2	[3.99077248 59.53602181]
C'_H guess #3	[3.99077248 59.53602181]
C'_H guess #4	[3.99077248 59.53602181]
C'_H bounds #1	[1.99538624 71.44322617]
C'_H bounds #2	[1.99538624 71.44322617]
C'_H bounds #3	[1.99538624 71.44322617]
C'_H bounds #4	[1.99538624 71.44322617]
Trials	1000
LFER Solver	SLSQP
maxiter (LFER solver)	1000
van't Hoff Solver	SLSQP
maxiter (van't Hoff solver)	1000
ftol (SLSQP)	1e-07
xtol (trust-constr)	1e-07
gtol (trust-constr)	1e-07
Seed	None

Questions or Comments?

Please reach out through the pyDMS GitHub page (github.com/SmithLabMIT/pyDMS) or via email at smithlab_software.mit.edu.

Citation

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